Grogoni doc 3 June

Grogoni Product Requirements Document (PRD) & Development Blueprint v1.0

Document Status

Document Title: Grogoni Product Requirements Document (PRD) & Development Blueprint v1.0

Document Type: Product, Delivery and Engineering Planning

Companion Documents

- Document 1 — Grogoni Architecture v3.0
- Document 2 — Technical Architecture Specification v1.0
- Document 4 — Grogoniclaw Runtime Specification v1.0

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1. Purpose

This document converts the architecture into an executable development plan.

It defines:

- Product vision
- User personas
- Functional requirements
- Non-functional requirements
- Epics
- Features
- User stories
- Acceptance criteria
- Milestones
- Sprint roadmap
- Testing strategy
- Release strategy

- Success metrics

It is intended to answer one question:

What must be built, in what order, and how do we know it is complete?

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2. Product Vision

Grogoni is an AI-native business operating system designed for low-resource markets.

It enables small businesses to run commerce using conversation instead of complex software while remaining usable on low-end Android smartphones with intermittent internet connectivity.

The platform should become the primary operating system for:

- Retail shops
- Agribusinesses
- Wholesalers
- Distributors
- Delivery businesses
- Service providers
- Informal merchants

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3. Product Principles

Every feature must satisfy the following principles.

3.1 Mobile First

Every workflow begins on a phone.

Desktop is optional.

3.2 Chat First

Every operation must be executable from conversation.

Examples:

Add customer

Record payment

Create invoice

Import products

Check debt

Send reminder

3.3 Offline First

Every critical workflow should continue working without internet.

3.4 AI Assisted

AI accelerates work but never silently performs irreversible business operations.

3.5 Structured Data

Every conversational action should produce structured business records.

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4. User Personas

Owner

Responsibilities

- Configure business
- View reports
- Manage staff
- Manage subscriptions
- Approve sensitive actions

Pain Points

- Paper records
- Lost inventory
- Credit tracking
- Tax preparation

Sales Agent

Responsibilities

- Sell
- Create invoices
- Track deliveries

Needs

- Fast product lookup
- Customer history
- Offline operation

Cashier

Responsibilities

- Receive payments
- Issue receipts
- Balance cash

Supplier

Responsibilities

- Deliver products
- Receive purchase requests
- Track orders

Customer

Responsibilities

- View invoices
- Pay
- Request quotations
- Confirm deliveries

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5. Product Goals

Launch Goals

- Phone onboarding under 3 minutes
- First invoice within 5 minutes
- Offline product management
- AI-powered conversational workflow

Growth Goals

- Marketplace
- Industry templates
- AI skills
- Digital twin
- Cross-border commerce

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6. Functional Requirements

The MVP shall provide:

Authentication

- Phone OTP
- Email OTP

Business

- Products
- Customers
- Suppliers
- Inventory
- Invoices
- Payments
- Credit

AI

- Conversational assistant
- Intent recognition
- Guided workflows

Device

- Camera
- File upload
- Contacts
- Notifications

Platform

- Offline mode
- Sync
- Audit logs

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7. Non-Functional Requirements

Performance

- Chat response < 2 s for local actions
- Invoice draft < 2 s
- App shell < 15 MB

Reliability

- Offline queue

- Retry-safe writes
- Conflict resolution

Security

- RBAC
- Encryption
- Audit logs

Scalability

- Multi-tenant
- Horizontal API scaling
- Event-driven workers

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8. Product Epics

Epic 1 — Account & Identity

Features

- OTP login
- Business creation
- Language selection
- Subscription plan
- Staff management

Success Criteria

User creates a business account in under 3 minutes.

Epic 2 — Business Core

Features

- Products
- Customers
- Suppliers
- Inventory

- Invoices
- Payments

Success Criteria

Business can replace a paper ledger with Grogoni.

Epic 3 — AI Assistant

Features

- Chat
- Intent routing
- Guided forms
- Tool execution
- Context awareness

Success Criteria

User completes common workflows entirely through conversation.

Epic 4 — Offline Platform

Features

- Local database
- Event queue
- Delta sync
- Conflict handling

Success Criteria

Business continues operating without connectivity.

Epic 5 — Document Import

Features

- Spreadsheet import
- PDF import
- Receipt extraction
- Preview and confirmation

Success Criteria

Existing business records migrate with minimal manual entry.

Epic 6 — Payments

Features

- Manual payment
- Mobile money integration
- Invoice payment status
- Debt tracking

Success Criteria

Business can reconcile invoices with payments.

Epic 7 — Logistics

Features

- Deliveries
- Vehicles
- Drivers
- Routes

Success Criteria

Orders can be tracked from dispatch to delivery.

Epic 8 — Marketplace

Features

- Skill installation
- Templates
- Industry packs
- Reviews
- Updates

Success Criteria

Third parties extend the platform safely.

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9. User Stories

Example

Story

As a shop owner

I want to create an invoice by chatting

So that I do not need to navigate forms.

Acceptance Criteria

- Customer found or created
- Products selectable
- Totals calculated
- Tax calculated
- Invoice saved
- Confirmation shown

Story

As a sales agent

I want to search products offline

So that I can continue selling when internet fails.

Acceptance Criteria

- Local search available
- Cached inventory shown
- Sync occurs later

Story

As a customer

I want to receive a payment link

So that I can pay remotely.

Acceptance Criteria

- Link generated
- Invoice updated after payment
- Receipt available

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10. Product Backlog

Priority 1

- Login
- Products
- Customers
- Invoices
- Offline storage

Priority 2

- Payments
- Inventory
- Reports

Priority 3

- Document import
- Marketplace
- AI enhancements

Priority 4

- Logistics
- Digital twin
- Forecasting

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11. Sprint Roadmap

Sprint 1

- Authentication
- Business creation
- Chat shell

Sprint 2

- Products
- Customers

Sprint 3

- Invoices
- Inventory

Sprint 4

- Offline database
- Sync engine

Sprint 5

- AI assistant

Sprint 6

- Payments

Sprint 7

- Documents

Sprint 8

- Reports

Sprint 9

- Logistics

Sprint 10

- Marketplace

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12. Definition of Done

A feature is complete only when:

- Code implemented
- Unit tests pass
- Integration tests pass
- Offline tested
- AI safety reviewed
- Accessibility reviewed
- Documentation updated
- Performance verified
- Acceptance criteria met

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13. QA Strategy

Testing includes:

- Unit testing
- Integration testing
- End-to-end testing
- Offline testing
- Device compatibility testing
- AI behaviour evaluation
- Security testing
- Load testing

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14. Release Plan

Alpha

Internal team only.

Focus:

- Architecture validation
- Core CRUD
- Basic AI

Beta

Selected merchants.

Focus:

- Offline reliability
- Sync stability
- Payment testing
- UX refinement

Public Launch

Focus:

- Marketplace
- Industry packs
- Customer portal
- Regional rollout

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15. Success Metrics

Technical

- Crash-free sessions
- Sync success rate
- AI task completion rate
- API latency
- Offline reliability

Business

- Daily active businesses
- Invoices created
- Payments processed
- Customer retention
- Marketplace adoption

User Experience

- Time to first invoice
- Time to first product
- AI task success
- User satisfaction
- Feature adoption

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16. Team Structure

Core Team

- Product Manager
- Technical Architect
- Engineering Lead
- AI Lead
- Mobile Engineer
- Backend Engineer
- Frontend Engineer
- QA Engineer
- UX Designer
- DevOps Engineer

Future Roles

- Marketplace Manager
- Developer Relations
- Customer Success
- Compliance Specialist
- Data Analyst

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17. Risks and Mitigation

Technical Risks

- Low-end device limitations
- AI latency
- Offline conflicts

Mitigation

- Progressive enhancement
- Local-first execution
- Event-driven sync

Business Risks

- Merchant trust
- Payment reliability
- Regulatory changes

Mitigation

- Verification tiers
- Audit trails
- Configurable compliance

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18. Go-to-Market Plan

Phase 1

Target independent retailers and agribusinesses in Kenya.

Phase 2

Expand to wholesalers, distributors, and delivery businesses.

Phase 3

Roll out regional support with localized tax, payments, and language packs.

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19. Long-Term Product Evolution

Future capabilities include:

- Multi-agent collaboration
- Autonomous procurement suggestions
- Predictive inventory management
- Demand forecasting
- Cross-business supply chain optimisation
- Federated learning for local models
- Voice-first operation
- AI-powered business advisory

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20. Document Relationships

The complete documentation suite is organised as follows:

- **Document 1 – Grogoni Architecture v3.0:** Defines the vision, architecture, data model, platform structure, and system behaviour.
- **Document 2 – Technical Architecture Specification v1.0:** Defines implementation details, repository layout, APIs, infrastructure, schemas, deployment, and engineering standards.
- **Document 3 – Product Requirements Document & Development Blueprint v1.0:** Defines product scope, delivery roadmap, epics, user stories, milestones, testing, releases, and success metrics.
- **Document 4 – Grogoniclaw Runtime Specification v1.0:** Defines the AI runtime internals, including planning, orchestration, memory, model routing, tool execution, verification, skills, and runtime lifecycle.

Together, these four documents provide a complete blueprint for designing, building, deploying, **and evolving Grogoni from MVP to a scalable AI-native business operating system.**